Amaury Baret

Doctoral Researcher & Teaching Assistant — Condensed-Matter Physics — Percolation, Transport & Optics of Stochastic Media

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RESEARCH INTERESTS

- Transport, percolation, and non-equilibrium phenomena in stochastic and disordered metallic nanowire networks
- Optical response of metallic nanowire networks: Mie scattering, transfer-matrix modelling, effective-medium theory
- Thermochromic and low-emissivity stacks (VO_2, silver nanowires) for autonomous thermal regulation and passive radiative cooling
- High-performance numerical simulation of charge transport and percolation using graph theory
- Integration of functional oxides with transparent conducting materials by RF magnetron sputtering

EDUCATION

Ph.D. in Physics — Doctoral Researcher & Teaching Assistant *2023 — present*

University of Liège, Department of Physics, SPIN Laboratory · Liège, Belgium

Dissertation (in progress) — Metallic nanowire networks as active transparent electrodes for thermochromic devices. Transport, percolation, and optical response of disordered metallic nanowire networks, with applications to smart-window, low-emissivity, and VO_2-based thermochromic stacks.

Supervisor: Prof. Ngoc Duy Nguyen.

M.Sc. in Condensed-Matter Physics — *summa cum laude* *2020 — 2022*

University of Liège, Department of Physics · Liège, Belgium

Master's thesis — Numerical investigation of low-density metallic nanowire networks as a cure for defective transparent conducting materials. Computational study of sparse silver-nanowire overlays restoring electrical percolation through damaged transparent conductors; precursor to the Nanoscale paper on bridge percolation.

B.Sc. in Physics — *magna cum laude* *2017 — 2020*

University of Liège, Faculty of Sciences · Liège, Belgium

Core physics curriculum.

INTERNATIONAL COLLABORATIONS & RESEARCH STAYS

M-ERA.NET — INSTEAD *2023 — present*

LMGP (Université Grenoble Alpes, FR) · ICMCB (Université de Bordeaux, FR) · Ankara University (TR)

European partnership on silver-nanowire-based transparent conductors and their integration with functional oxides. Led several work-package contributions on optical characterisation and device integration. Carried out experimental research stays in Grenoble (LMGP) and Bordeaux (ICMCB) for fabrication and characterisation campaigns.

PUBLICATIONS & PREPRINTS

Peer-reviewed articles and preprints, reverse chronological. Selected publications marked ★. Underlined author indicates the candidate. A full list is maintained on ORCID and Google Scholar.

[P1] A. Baret, A. Khan, S. Akin, L. Teulé-Gay, D. Bellet, A. Rougier, N. D. Nguyen (2026). *Challenges and mitigation pathways in coating silver nanowire networks with metallic oxides by RF magnetron sputtering*. *arXiv preprint*, 2604.09372 [PREPRINT]. doi:10.48550/arXiv.2604.09372

[P2] A. Baret, J. Baumgarten, F. Balty, F. Rabecki, J. Brisbois, B. Zheng, D. Bellet, N. D. Nguyen (2025). *The refractive index of silver nanowire networks: a heuristic approach to the foundations of the optical constants, from experiment to theory*. *Discover Nano*, 20, 131. doi:10.1186/s11671-025-04312-9

[P3] T. Ratz, E. Fourneau, N. Sliti, C. Malherbe, A. Baret, B. Vertruyen, A. Silhanek, N. D. Nguyen (2025). *Correlation between material properties, crystalline transitions, and point defects in RF sputtered (N,Mg)-doped copper oxide thin films*. *ACS Applied Electronic Materials*, 7(2). doi:10.1021/acsaelm.4c01396

[P4] ★ A. Baret (2025). *Reconnecting the Fractured: Nanowire Networks and the Physics of Bridge Percolation*. *Bulletin de la Société Royale des Sciences de Liège*, 94(1), 80–101. doi:10.25518/0037-9565.12531

> Sole-author pedagogical paper; basis for the 2025 Prize of the Royal Society of Sciences of Liège.

[P5] ★ A. Baret, A. Khan, A. Rougier, D. Bellet, N. D. Nguyen (2025). *Low-emissivity fine-tuning of efficient VO2-based thermochromic stacks with silver nanowire networks*. *RSC Applied Interfaces*, 2(1), 94–103. doi:10.1039/d4lf00234b

[P6] F. Balty, A. Baret, A. Silhanek, N. D. Nguyen (2024). *Insight into the morphological instability of metallic nanowires under thermal stress*. *Journal of Colloid and Interface Science*. doi:10.1016/j.jcis.2024.06.074

[P7] ★ A. Baret, L. Bardet, D. Oser, D. Langley, F. Balty, D. Bellet, N. D. Nguyen (2024). *Bridge percolation: electrical connectivity of discontinued conducting slabs by metallic nanowires*. *Nanoscale*, 16, 8361–8368. doi:10.1039/d3nr05850f

Thesis

[P1] A. Baret (2022). *Numerical investigation of low-density metallic nanowire networks as a cure for defective transparent conducting materials*. *M.Sc. Thesis — University of Liège*, *summa cum laude*. matheo.uliege.be/handle/2268.2/14793

CONFERENCE TALKS & POSTERS

May 2026 — CONTRIBUTED TALK + POSTER (FORTHCOMING)
"E-MRS Spring Meeting 2026, Symposium W"
Strasbourg, France

Oct 2025 — CONTRIBUTED TALK
"Heuristic approach to the fundamental optical constants of silver nanowire networks: experiments and theory".
TCM-TOEO 2025 — Rethymno, Greece

Oct 2025 — POSTER
"Thermal emissivity of silver nanowire networks: a characterization tool for instability studies".
TCM-TOEO 2025 — Rethymno, Greece

Oct 2025 — POSTER (CO-AUTHOR)
"Crystalline transitions and point defects in (N, Mg)-doped copper oxide thin films deposited by radiofrequency magnetron sputtering".
TCM-TOEO 2025 — Rethymno, Greece

Sep 2025 POSTER
"Crystalline phase transitions and point defects in N- and Mg-doped copper oxide thin films deposited by magnetron sputtering".
Belgian Crystallography Symposium — 13 — Leuven, Belgium

Jun 2025 CONTRIBUTED TALK
"Optical constants of silver nanowire networks: theory and experiments".
Materials Today Conference 2025 — Sitges, Spain

Jun 2025 POSTER
"On the thermal emissivity of highly efficient VO2-based thermochromic stacks coated with silver nanowire networks".
Materials Today Conference 2025 — Sitges, Spain

May 2025 CONTRIBUTED TALK
"Optical constants of silver nanowire networks: theory and experiments".
E-MRS Spring Meeting 2025, Symposium F — Strasbourg, France

May 2024 CONTRIBUTED TALK
"Low-emissivity fine-tuning of efficient VO2-based thermochromic stacks via silver nanowire networks".
E-MRS Spring Meeting 2024, Symposium A — Strasbourg, France

Jun 2023 POSTER
"Bridge percolation: electrical connectivity of discontinued conducting slabs by metallic nanowires".
E-MRS Spring Meeting 2023 — Strasbourg, France

Oct 2022 POSTER
"Composite transparent conducting material using metallic nanowire networks on a flexible substrate".
TCM-TOEO 2022 — Hersonissos, Greece

AWARDS, HONOURS & DISTINCTIONS

Research & academic

2025 **Prix annuel de la Société Royale des Sciences de Liège**
Annual prize of the Royal Society of Sciences of Liège, recognising broadly accessible scientific writing (sole-author paper on bridge percolation in nanowire networks).

2025 **Best Poster Award — Materials Today Conference 2025**
Sitges, Spain. For work on the thermal emissivity of VO2-based thermochromic stacks coated with silver nanowire networks.

2025 **Best Oral Presentation Award — ULiège Ph.D. Day 2025**
University of Liège annual doctoral symposium, across all disciplines.

Additional distinctions

2025 **BNP Paribas Fortis & UCM Startup Prizes — Athena**
Co-founder of Athena, a startup applying optimisation algorithms to medical scheduling; recognised by BNP Paribas Fortis and the UCM for the computational approach and entrepreneurial execution.

2024 **2nd place — Odoo Hackathon 2024**
Web-development and rapid-prototyping competition hosted by Odoo.

2024 **Nominee & Best Presentation — Japan–Belgium Royal Mission Game Jam**
Team 16, Verdant. Recognised for communication of design intent under time pressure.

2023 **Grand Prize & Best Presentation — Citizens of Wallonia Hackathon**
Largest hackathon in Wallonia. Drone-mapping tool for risk-aware inter-hospital organ transport, built on Proximus API data with graph-theoretic routing.

TEACHING & SUPERVISION

Courses (University of Liège)

Introduction to Electromagnetism (PHYS-1986) *2023 — present*
Teaching Assistant · First-year undergraduate (Physics)
Weekly tutorial and exercise sessions.

Introduction to Modern Physics & Quantum Mechanics (PHYS-2026) *2023 — present*
Teaching Assistant · Second-year undergraduate (Engineering)
Tutorials in special relativity, early quantum theory, and introductory quantum mechanics.

Electronics & Experimental Techniques (PHYS-0092) *2023 — present*
Lab Instructor · Third-year undergraduate (Physics)
Laboratory supervision and tutoring on analog and digital electronics and experimental methodology.

Supervision

Co-supervisor — 4 M.Sc. theses *2023 — present*
Three theses on transport in metallic nanowire networks; one on thermochromic materials.

Supervisor — 8 undergraduate interns *2023 — present*
Physics and engineering students on experimental projects; 5 of 8 have since begun a Ph.D.

TECHNICAL SKILLS

Experimental

Thin-film deposition	RF magnetron sputtering, spray-coating, spin-coating
Structural characterisation	SEM, XRD, profilometry
Optical & electrical metrology	UV–Vis–NIR & IR spectrophotometry, emissivity measurement, DC sheet-resistance characterisation

Computational

Languages	Python (advanced), C / C++ (advanced), Java, SQL, Mathematica
High-performance computing	MPI, OpenMP, SLURM-scheduled clusters
Simulation & modelling	COMSOL Multiphysics (FEM), Mie scattering, transfer-matrix method, percolation & graph-theory solvers (custom)
Scientific workflow	Git, LaTeX, Linux, numerical analysis, data visualisation

LANGUAGES

French	Native
English	Fluent (C2)
Dutch	Intermediate (B1)

REFERENCES

Available upon request.